

Proofs and Problem Solving

Week 3: Bounds and Limits

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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5. Upper Bounds and Least Upper Bounds

For the next two weeks, we'll learn about the basics of *analysis*. This branch of mathematics studies limits and approximations, and you may have already met these when studying calculus. If you end up enjoying this topic, you might also like the following courses in subsequent years:

- Fundamentals of Pure Mathematics, 2nd year.
- Honours Analysis, 3rd year.
- Linear Analysis, Essentials in Analysis and Probability, Functional Analysis, Fourier Analysis, 4th year.

Why care? Firstly, a lot of what we now call analysis was not very rigorous prior to the 1800s. The mathematician Niels Abel went so far as to say the following about the state of analysis at the time:

*I shall devote all my efforts to bring light into the immense obscurity that today reigns in Analysis. It so lacks any plan or system, that one is really astonished that so many people devote themselves to it – and, still worse, it is absolutely devoid of any rigour.*¹

To give an idea about what he was talking about, Leibniz (one of the founders, with Newton, of modern calculus) believed at one time that

$$\sum_{n=0}^{\infty} (-1)^n = \frac{1}{2},$$

but with modern rigour we now understand that this is sadly not true. In fact, this sum is meaningless, as we shall see later. This infinite sum is known as Grandi's series. Making analysis rigorous was a big problem since it is the foundation of calculus: many of the computations you perform in calculus, such as taking derivatives and evaluating integrals, in fact rely on analysis "under the bonnet". More shockingly, we need tools from analysis

¹Niels Abel, *Oeuvres* (1826).

even to just take n th roots, as we shall see later as well. Before getting into limits, however, we will need to discuss upper and lower bounds for sets, and to introduce a final axiom for the real numbers.

5.1 Upper and Lower Bounds

Definition 5.1 Let A be a subset of $\mathbb{R}$.

- An *upper bound* for A is a real number M such that for all $x \in A$, we have $x \leq M$. We say that A is *bounded above* if there exists an upper bound for A , and *unbounded above* otherwise.
- A *lower bound* for A is a real number m such that for all $x \in A$, we have $m \leq x$. We say that A is *bounded below* if there exists a lower bound for A , and *unbounded below* otherwise.
- We say A is *bounded* if it is both bounded above and bounded below.

■ **Exercise 5.1** Show that A is bounded if and only if there is some $t \geq 0$ such that $|x| \leq t$ for all $x \in A$.

Recall that we write intervals using the notation

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} \mid a < x < b\},$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}, \quad (a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}.$$

For unbounded intervals, we write

$$[a, \infty) = \{x \in \mathbb{R} \mid a \leq x\}, \quad (a, \infty) = \{x \in \mathbb{R} \mid a < x\},$$

$$(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\}, \quad (-\infty, b) = \{x \in \mathbb{R} \mid x < b\}.$$

Example 5.1 (a) Let $A = [0, 3)$. Then A is bounded above since 3 is an upper bound; A is bounded below since 0 is a lower bound.

(b) Let $A = \{t + \frac{1}{t} : t > 0\}$. Then A is bounded below since 0 is a lower bound. (We could have also shown it was bounded below by using the AM-GM inequality: for $t > 0$,

$$\frac{t + 1/t}{2} \geq \sqrt{t \cdot \frac{1}{t}} = 1,$$

so 2 is also a lower bound for A .) But A is not bounded above, since, for any $M \in \mathbb{R}$ we can find $t > 0$ such that $t + \frac{1}{t} > M$: if $M \leq 0$ then any $t > 0$ works; if $M > 0$ then $t = M$ works since $M + \frac{1}{M} \in A$ is greater than M .

(c) Let $A = \{(-1)^n(1 - \frac{1}{n}) \mid n \in \mathbb{N}\} = \{0, \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \frac{5}{6}, \dots\}$. Then A is bounded: 1 is an upper bound since for all n :

$$(-1)^n \left(1 - \frac{1}{n}\right) \leq 1 - \frac{1}{n} < 1$$

and one can show similarly -1 is a lower bound.

- (d) Let $A = \{\sin n \mid n \in \mathbb{N}\} = \{\sin 1, \sin 2, \sin 3, \dots\}$. Because $-1 \leq \sin x \leq 1$ for all x , we have $A \subseteq [-1, 1]$, so A is bounded.

5.2 Least Upper Bounds and Greatest Lower Bounds

A subset of $\mathbb{R}$ that is bounded above will have many upper bounds: if M is an upper bound for a set A , so is every number greater than M . It is often useful to know what is the best or *least* upper bound for a set.

Definition 5.2 Given a subset $A \subseteq \mathbb{R}$, a number L is a *least upper bound* (LUB) or *supremum* for A if

- (a) L is an upper bound for A , that is, $x \leq L$ for all $x \in A$; and
- (b) $L \leq M$ for every upper bound M of A .

Similarly, we say ℓ is a *greatest lower bound* (GLB) or *infimum* for A if

- (a) ℓ is a lower bound for A , that is, $x \geq \ell$ for all $x \in A$; and
- (b) $\ell \geq m$ for every lower bound m of A .

If an LUB exists for a set A , we denote it by $\text{LUB}(A)$. Similarly, we denote a GLB for A by $\text{GLB}(A)$.

Condition (b) in the definition of LUB says that among all upper bounds for A , the upper bound L is the least or smallest one. Another way to say (b) is that no number smaller than L is an upper bound: for all $t < L$, there exists $x \in A$ such that $x > t$.

If L and L' are both least upper bounds for A then $L \leq L'$ and $L' \leq L$ by (b), so $L = L'$. Hence a subset of $\mathbb{R}$ can have at most one least upper bound. This allows us to speak of *the* least upper bound of A (if it has one at all), rather than just *a* least upper bound of A .

When proving that a number is the LUB for some set, we follow a two-step process of verifying conditions (a) and (b) in the definition.

Example 5.2 We claim that the least upper bound of $(-\infty, 1)$ is 1.

- (a) 1 is an upper bound for the set, by definition of $(-\infty, 1)$.
- (b) Let M be an upper bound for $(-\infty, 1)$. Suppose for a contradiction that $M < 1$. Then $\frac{1+M}{2} < 1$, so $\frac{1+M}{2} \in (-\infty, 1)$. But $M < 1$ implies $\frac{1+M}{2} > M$, contradicting M being an upper bound for $(-\infty, 1)$. Hence $M \geq 1$.

Example 5.3 Let $S = \{x \in \mathbb{Q} \mid x < 0\}$. We claim that $\text{LUB}(S) = 0$.

- (a) Since every $x \in S$ is negative by definition, 0 is an upper bound.
- (b) Let M be an upper bound for S . Suppose for a contradiction that $M < 0$. Recall from Week 1 (Theorem 2.4) that whenever $a, b \in \mathbb{R}$ with $a < b$, the interval (a, b) contains at least one rational number. So there is some $x \in \mathbb{Q}$ such that $M < x < 0$. Then $x \in S$ with $x > M$, contradicting M being an upper bound for S . Hence $M \geq 0$.

Example 5.4 Let $A = \{\frac{1}{n} \mid n \in \mathbb{N}\}$. We claim that $\text{GLB}(A) = 0$.

Again, we verify the two conditions of being a GLB:

- (a) Since $\frac{1}{n} > 0$ for all $n \in \mathbb{N}$, it follows that 0 is a lower bound for A .
- (b) Let m be a lower bound for A . Suppose for a contradiction that $m > 0$. By the Archimedean property, there is some $n \in \mathbb{N}$ such that $n > 1/m$. Then $1/n \in A$ with $1/n < m$, contradicting m being a lower bound for A . Hence $m \leq 0$.

Given that a maximum or minimum of a set need not exist, why should the least upper

bound (which is just the minimum of all upper bounds) exist at all? It turns out that this is an important axiom for the real numbers which we shall need to assume, and it cannot be derived from any of the axioms we have assumed so far:

Completeness Axiom for the real numbers: Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

■ **Exercise 5.2** Why does the completeness axiom imply that every nonempty subset of $\mathbb{R}$ that is bounded below has a GLB?

An important consequence of the Completeness Axiom is the existence of n th roots of positive real numbers as described in Proposition 2.7. We will establish this as Theorem 6.5. Another very important consequence of the Completeness Axiom for the real numbers is the Archimedean property, mentioned in Week 1:

Theorem 5.1 The Archimedean property holds. That is, for any $x, y \in \mathbb{R}$ with $x, y > 0$, there is some $n \in \mathbb{N}$ such that $ny > x$.

Proof. Assume for a contradiction that the theorem is false. Then for some positive $x, y \in \mathbb{R}$, every $n \in \mathbb{N}$ satisfies $ny \leq x$. Hence x/y is an upper bound for $\mathbb{N}$, so $\mathbb{N}$ is bounded above. By the Completeness Axiom, $\mathbb{N}$ has a least upper bound L . For all $n \in \mathbb{N}$, we have $n+1 \in \mathbb{N}$ too, so $n+1 \leq L$. Hence $n \leq L-1$ for all $n \in \mathbb{N}$. But this means that $L-1$ is an upper bound for $\mathbb{N}$, contradicting L being the *least* upper bound. ■

5.3 Exercises

The exercises in Liebeck's book relevant to this section are in Chapter 22.

■ **Exercise 5.3** Let $A, B \subseteq \mathbb{R}$ be nonempty sets that are bounded above. For each statement below, either prove it or provide a counterexample.

- If A is bounded above, then $\text{LUB}(-A) = -\text{LUB}(A)$, where $-A = \{-x \mid x \in A\}$.
- If A is bounded above, then $-A$ is bounded below and $\text{GLB}(-A) = -\text{LUB}(A)$.
- If $c \in \mathbb{R}$, then $\text{LUB}(c+A) = c + \text{LUB}(A)$ where $c+A = \{c+x \mid x \in A\}$.
- $\text{LUB}(A) = \max(A)$.
- If $c > 0$, then $\text{LUB}(cA) = c\text{LUB}(A)$ where $cA = \{cx \mid x \in A\}$.
- $\text{LUB}(AB) = \text{LUB}(A) \cdot \text{LUB}(B)$ where $AB = \{xy \mid x \in A, y \in B\}$.
- if $A \subseteq B$, then $\text{LUB}(A) \leq \text{LUB}(B)$.

■ **Exercise 5.4** Suppose A and B are bounded above. Show that $\text{LUB}(A+B) = \text{LUB}(A) + \text{LUB}(B)$, where $A+B = \{x+y \mid x \in A, y \in B\}$.

■ **Exercise 5.5** Determine which of the following sets are bounded or unbounded above.

- $A = \{x \in \mathbb{R} \mid x^2 - 3x + 2 < 2\}$.
- $A = \{x \in \mathbb{R} \mid \cos x < 1/2\}$. [You may assume any familiar properties of the cosine function.]
- $A = \{n^2 - n \mid n \in \mathbb{N}\}$.
- $A = \{ab \mid a, b \in \mathbb{R}, a+b=1\}$.

■ **Exercise 5.6** Find the least upper bounds for the following sets:

- (a) $\{\frac{1}{n} \mid n \in \mathbb{N}\}$.
- (b) $\{-\frac{1}{n^2} \mid n \in \mathbb{N}\}$.
- (c) $\{2 - x^2 \mid x \in \mathbb{R} \setminus \{0\}\}$.
- (d) $\{x \in \mathbb{Q} : x^2 < 2\}$.

6. Limits

6.1 Motivation and the ϵ - N definition of a limit

How does a computer compute $\sqrt{51}$? A computer can't possibly memorise all possible square roots, so it must compute it from scratch. Moreover, it can't possibly compute the entire decimal expansion (next week we will see that, because $\sqrt{51}$ is irrational, the decimal expansion never repeats), so it can only come up with an approximation. Thus, your computer requires an algorithm for computing $\sqrt{51}$ (or any root) to any degree of accuracy. But before implementing such an algorithm, we need to *justify* that it can indeed approximate $\sqrt{51}$ to any degree of accuracy. That is, if my algorithm spits out a sequence of numbers $x_1, x_2, \dots$, and I want an approximation that is within 10^{-10} of $\sqrt{51}$, I need to show that for all large enough n , x_n is at most 10^{-10} away from $\sqrt{51}$, and I need this to happen for *every* degree of accuracy $\epsilon > 0$ that I may require (not just 10^{-10}). All this provides a good reason to study limits of sequences.

Here, we define sequences and then their limits.

Definition 6.1 For $k \in \mathbb{Z}$, $(a_n)_{n=k}^{\infty}$ is defined to be a real-valued sequence beginning at index k if, for each integer $n \geq k$, the quantity a_n is a real number. For each $n \geq k$, the n th term of the sequence is defined to be a_n .

A sequence $(a_n)_{n=1}^{\infty}$ can also be thought of as an ordered list. For example, for the sequence $(\frac{1}{n})_{n=1}^{\infty}$, the first terms is $a_1 = 1/1 = 1$, the second term is $a_2 = 1/2$, and so on, so that the sequence is the ordered list beginning $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ and continuing on forever in this way. Often, we use the notation (a_n) for simplicity. Note that, when writing a sequence, we put the terms in parentheses (round brackets "(" and ")"), such as $(b_i)_{i=n}^{\infty}$, to indicate the sequence of terms is ordered, in contrast to a set $\{b_i : i \geq n\}$, which is not ordered and for which we use what typographers call braces (curly brackets "{" and "}"). Sometimes in mathematics, we also want to consider sequences taking values not in $\mathbb{R}$ but in a set of vectors, a set of matrices, or other sets.

There are many examples. We already saw the sequence starting at $n = 1$, where the n th term is $\frac{1}{n}$; this sequence can't be considered as a sequence starting at $n = 0$, since $\frac{1}{n}$

is not defined for $n = 0$. Examples of sequences that could start at $n = 0$ would be those where the n th term is $a_n = n^2 + 3n$, $b_n = e^n$, or $c_n = \frac{n^3 + 4n}{n^2 + 2n + 3}$. For these examples, there is a simple formula for the sequence, but other sequences might be more complicated. For example, we can define a sequence (e_n) where e_n is the real number with the first n digits of π ; here the first 5 terms would be 3, 3.1, 3.14, 3.141, and 3.1415. Another example would be $(f_n)_{n=1}^\infty$ where $f_n = 1$ if n is prime and $f_n = 0$ if n is not prime. Other sequences might be determined in even more complicated ways.

For a sequence, whether the terms defined by a simple formula or something more complicated, we can now define the notion of limit.

Definition 6.2 — The ϵ - N definition of a limit. Let $(x_n)_{n=1}^\infty$ be a sequence of real numbers $x_1, x_2, \dots$ and let $L \in \mathbb{R}$. We say that x_n *converges* to L , or L is the *limit* of the sequence (x_n) , if for all $\epsilon > 0$, there exists a natural number N such that for all $n > N$,

$$|x_n - L| < \epsilon.$$

When x_n converges to L , we write $x_n \rightarrow L$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = L$.

An equivalent way of saying $|x_n - L| < \epsilon$ is

$$L - \epsilon < x_n < L + \epsilon$$

(Exercise 6.1).

Definition 6.3 We say that a sequence (a_n) is *convergent* if for some $L \in \mathbb{R}$, it converges to L .

Exercise. Give an example of a convergent sequence. Give an example of a non-convergent sequence.

Notice that here we are working with the notion of a **sequence** $(x_n)_{n=1}^\infty$ – this is just an *ordered* list of real numbers x_n , indexed by the natural numbers $\mathbb{N}$. Or, if you prefer, you can think of a sequence as a function $x : \mathbb{N} \rightarrow \mathbb{R}$ with domain $\mathbb{N}$ and codomain $\mathbb{R}$.

It is important to recognise that in Definition 6.2, we expect N to depend on ϵ .

Example 6.1 Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

We need to show that for all $\epsilon > 0$ there is N such that $n > N$ implies $|\frac{1}{n} - 0| < \epsilon$, which is the same as $\frac{1}{n} < \epsilon$. This is equivalent to finding N such that $n > N$ implies $n > \frac{1}{\epsilon}$. By the Archimedean property, we can pick an integer $N > \frac{1}{\epsilon}$. Then for all $n > N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \frac{1}{N} < \epsilon.$$



Note on establishing limits: Don't expect a nice looking proof to fall into your lap immediately when trying to establish a certain limit. It usually requires some rough work first, which you can then use to help you write your proof.

Example 6.2 Show that $\lim_{n \rightarrow \infty} \frac{5n^2 + 2n}{n^2 + 4} = 5$.

Rough work: We are going to need to show that $\frac{5n^2 + 2n}{n^2 + 4} - 5$ is small when n is large,

so it first makes sense to simplify this expression algebraically to see if we can identify when it might be small. Doing this,

$$\frac{5n^2+2n}{n^2+4} - 5 = \frac{2n-20}{n^2+4},$$

and therefore, using the triangle inequality and the fact that $4 \geq 0$,

$$\left| \frac{5n^2+2n}{n^2+4} - 5 \right| \leq \left| \frac{2n-20}{n^2+4} \right| \leq \frac{2n}{n^2+4} + \frac{20}{n^2+4} \leq \frac{2}{n} + \frac{20}{n^2}.$$

This is looking good because all we need to do to make this small is to make both $2/n$ and $20/n^2$ small, and this is very reasonable. Now we bring in the $\epsilon > 0$. If n is such that both $2/n < \epsilon/2$ and $20/n^2 < \epsilon/2$, we can deduce that

$$\left| \frac{5n^2+2n}{n^2+4} - 5 \right| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

which is exactly what we need. What do we need n to satisfy for this to be true? Well, for $\frac{2}{n} < \frac{\epsilon}{2}$ we need $n > \frac{4}{\epsilon}$, and for $\frac{20}{n^2} < \frac{\epsilon}{2}$ we need $n > \sqrt{\frac{40}{\epsilon}}$. To ensure both of these hold, we need n to satisfy $n > \max \left\{ \frac{4}{\epsilon}, \sqrt{\frac{40}{\epsilon}} \right\}$. So we take an integer $N \geq \max \left\{ \frac{4}{\epsilon}, \sqrt{\frac{40}{\epsilon}} \right\}$.

(All the above was just rough work to help me formulate and write my formal proof. Below is what you want to do when writing your work for marking; don't write the rough work. It's fine not to show how you chose your N , so long as it works. And there is no "single answer" for N : if N works, then so does N' for any $N' > N$.)

Now for the actual proof:

Claim: $\lim_{n \rightarrow \infty} \frac{5n^2+2n}{n^2+4} = 5$.

Proof. Let ϵ be any positive real number. We need to show that there is $N \in \mathbb{N}$ such that $n > N$ implies $\left| \frac{5n^2+2n}{n^2+4} - 5 \right| < \epsilon$. By the Archimedean property, we can choose an integer $N \geq \max \left\{ \frac{4}{\epsilon}, \sqrt{\frac{40}{\epsilon}} \right\}$. We claim that this N works. Indeed, if $n > N$, then

$$\left| \frac{5n^2+2n}{n^2+4} - 5 \right| \leq \left| \frac{2n-20}{n^2+4} \right| \leq \frac{2n}{n^2+4} + \frac{20}{n^2+4} \leq \frac{2}{n} + \frac{20}{n^2} < \frac{2}{N} + \frac{20}{N^2} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since we have shown we can find such an N no matter what $\epsilon > 0$ we started with, this proves the claim. ■

Example 6.3 Suppose $x_n \geq 0$ and $x_n \rightarrow x > 0$. Then $\sqrt{x_n} \rightarrow \sqrt{x}$.

Discussion: This is a little different from the previous example because now x_n isn't given in a concrete form, and all we know about it is that $x_n \rightarrow x$, or, more informally, $|x_n - x|$ is small for all large n . Nevertheless the conclusion, $\sqrt{x_n} \rightarrow \sqrt{x}$, is similar to what we had before, and the tactic of manipulating the expression $\sqrt{x_n} - \sqrt{x}$ algebraically,

and using some inequalities on it until it's apparent why it should also be small, is equally useful. Thus, we want to manipulate $|\sqrt{x_n} - \sqrt{x}|$ so that $|x_n - x|$ appears in the expression, since we hope that if we can make $|x_n - x|$ as small as we want, then we can make $|\sqrt{x_n} - \sqrt{x}|$ small as well.

A standard trick when working with roots is to multiply and divide by the conjugate, and using this trick here gives

$$|\sqrt{x_n} - \sqrt{x}| = \left| (\sqrt{x_n} - \sqrt{x}) \cdot \frac{\sqrt{x_n} + \sqrt{x}}{\sqrt{x_n} + \sqrt{x}} \right| = \left| \frac{x_n - x}{\sqrt{x_n} + \sqrt{x}} \right| = \frac{|x_n - x|}{\sqrt{x_n} + \sqrt{x}} \leq \frac{|x_n - x|}{\sqrt{x}}$$

where in the last step we used the facts that $\sqrt{x_n} \geq 0$ and $x > 0$. This is looking good because we should be able to make $\frac{|x_n - x|}{\sqrt{x}}$ small by taking n large enough. Indeed, now let's bring in $\epsilon > 0$. Since $x_n \rightarrow x$, we know that there is some $N \in \mathbb{N}$ such that whenever $n > N$, we have

$$|x_n - x| < \epsilon \sqrt{x}.$$

(This maybe requires a pause for thought. The definition of $x_n \rightarrow x$ talks about " $|x_n - x| < \epsilon$ " for all large n , it doesn't say anything about " $|x_n - x| < \epsilon \sqrt{x}$ ". But the definition of $x_n \rightarrow x$ applies to *every* positive ϵ – and thus it applies equally well to $\epsilon \sqrt{x}$ in place of ϵ . Remember that x is fixed in this discussion. The N coming out depends on x as well as ϵ , but this is not a problem, as x is fixed.) So if $n > N$, we have

$$|x_n - x| \leq \frac{|x_n - x|}{\sqrt{x}} < \frac{\epsilon \sqrt{x}}{\sqrt{x}} = \epsilon$$

Proof. We need to show that for any $\epsilon > 0$ there is $N \in \mathbb{R}$ such that $n > N$ implies $|\sqrt{x_n} - \sqrt{x}| < \epsilon$.

Let $\epsilon > 0$. Since $x_n \rightarrow x$, there is N such that $n > N$ implies $|x_n - x| < \epsilon \sqrt{x}$, and we take this N . We then have, for $n > N$,

$$\begin{aligned} |\sqrt{x_n} - \sqrt{x}| &= \left| (\sqrt{x_n} - \sqrt{x}) \cdot \frac{\sqrt{x_n} + \sqrt{x}}{\sqrt{x_n} + \sqrt{x}} \right| = \left| \frac{x_n - x}{\sqrt{x_n} + \sqrt{x}} \right| \\ &= \frac{|x_n - x|}{\sqrt{x_n} + \sqrt{x}} \leq \frac{|x_n - x|}{\sqrt{x}} < \frac{\epsilon \sqrt{x}}{\sqrt{x}} = \epsilon. \end{aligned}$$

■

Point to ponder: What happens if $x = 0$ in this example?

6.2 Rules for Limits

Definition 6.4 We say that a sequence (a_n) is *bounded* if the set of values $\{a_1, a_2, \dots\}$ is a bounded set, i.e. there are m, M such that $m \leq a_n \leq M$ for all n .

Proposition 6.1 Suppose the sequence (a_n) converges. Then it is bounded.

Proof. Let $L = \lim_{n \rightarrow \infty} a_n$. By the definition of a limit (with $\epsilon = 1$), there is N such that $n > N$ implies $|a_n - L| < 1$. (There was nothing special about our choice of $\epsilon = 1$ here; any

positive number will work in this argument.) Thus, for $n > N$,

$$|a_n| = |a_n - L + L| \leq |a_n - L| + |L| < 1 + |L|.$$

For $n \leq N$, we have $|a_n| \leq \max\{|a_1|, \dots, |a_N|\}$. Thus, for all n ,

$$|a_n| \leq \max\{1 + |L|, |a_1|, \dots, |a_N|\}.$$

By Exercise 5.1, the sequence is bounded. ■

Point to ponder: Does the converse to Proposition 6.1 hold?

Proposition 6.2 Suppose that (a_n) and (b_n) converge to a and b respectively. Then:

- (a) $a_n + b_n \rightarrow a + b$.
- (b) $a_n b_n \rightarrow ab$.
- (c) If $c \in \mathbb{R}$, then $ca_n \rightarrow ca$.
- (d) If $b \neq 0$ and $b_n \neq 0$ for all n , then $\frac{a_n}{b_n} \rightarrow \frac{a}{b}$.

Proof. (a) Let $\epsilon > 0$. We need to show there is N such that $n > N$ implies $|a_n + b_n - (a + b)| < \epsilon$. Note that by the triangle inequality,

$$|a_n + b_n - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b|.$$

We can make $|a_n + b_n - (a + b)| < \epsilon$ by making $|a_n - a|$ and $|b_n - b|$ both less than $\frac{\epsilon}{2}$. Since $a_n \rightarrow a$, we know that there is N_1 such that $n > N_1$ implies $|a_n - a| < \frac{\epsilon}{2}$. Similarly, since $b_n \rightarrow b$, we know that there is N_2 such that $n > N_2$ implies $|b_n - b| < \frac{\epsilon}{2}$. If we set $N = \max\{N_1, N_2\}$, then for all $n > N$,

$$|a_n + b_n - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

(b) Let $\epsilon > 0$. Note that

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - ab_n + ab_n - ab| = |(a_n - a)b_n + a(b_n - b)| \\ &\leq |(a_n - a)b_n| + |a(b_n - b)| = |a_n - a| \cdot |b_n| + |a| \cdot |b_n - b|. \end{aligned} \quad (6.2.1)$$

Thus, if we can pick N large enough that $n > N$ implies $|a_n - a| \cdot |b_n| < \frac{\epsilon}{2}$ and $|a| \cdot |b_n - b| < \frac{\epsilon}{2}$, then the above will imply $|a_n b_n - ab| < \epsilon$ and we'll be done. So let's focus on proving these two things.

Let's first find N_1 such that $n > N_1$ implies $|a_n - a| \cdot |b_n| < \frac{\epsilon}{2}$. Since b_n converges, it is bounded, and so there is M such that $|b_n| \leq M$ for all n . Thus

$$|a_n - a| \cdot |b_n| \leq M|a_n - a|.$$

Since $a_n \rightarrow a$, there is N_1 such that $n > N_1$ implies $|a_n - a| < \frac{\epsilon}{2M}$ and hence

$$|a_n - a| \cdot |b_n| \leq M|a_n - a| < M \cdot \frac{\epsilon}{2M} = \frac{\epsilon}{2}.$$

Now let's find N_2 such that $n > N_2$ implies $|a| \cdot |b_n - b| < \frac{\epsilon}{2}$. If $a = 0$, then this is always true and we can set $N_2 = 1$. If $a \neq 0$, then since $b_n \rightarrow b$, there is N_2 such that $n > N_2$ implies $|b_n - b| < \frac{\epsilon}{2|a|}$. Hence for all $n > N_2$,

$$|a| \cdot |b_n - b| < |a| \cdot \frac{\epsilon}{2|a|} = \frac{\epsilon}{2}.$$

Thus, if we set $N = \max\{N_1, N_2\}$, then for all $n > N$, (6.2.1) gives

$$|a_n b_n - ab| \leq |a_n - a| \cdot |b_n| + |a| \cdot |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

- (c) We leave this one as an exercise.
 (d) We claim that it suffices to show that if $b_n \rightarrow b$ and $b_n, b \neq 0$, then $b_n^{-1} \rightarrow b^{-1}$. Indeed, if this is the case, then by part (b),

$$\frac{a_n}{b_n} = a_n \cdot b_n^{-1} \rightarrow ab^{-1} = \frac{a}{b}.$$

So now let's prove that if $b_n \rightarrow b$ and $b_n, b \neq 0$, then $b_n^{-1} \rightarrow b^{-1}$.

Let $\epsilon > 0$. We need to find N such that $n > N$ implies $|b_n^{-1} - b^{-1}| < \epsilon$. Observe that

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \left| \frac{b - b_n}{bb_n} \right| = \frac{|b - b_n|}{|b| \cdot |b_n|}.$$

(The idea now: in the last expression, the numerator approaches 0 as $n \rightarrow \infty$, so if we can show that the denominator does not get too small, then the whole expression approaches 0.) Since $b_n \rightarrow b$, there is some N_1 such that for all $n > N_1$, we have $|b_n - b| < |b|/2$. But then by the reverse triangle inequality,

$$|b_n| = |b_n - b + b| \geq |b| - |b_n - b| > |b| - \frac{|b|}{2} = \frac{|b|}{2}.$$

Hence for all $n > N_1$,

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| < \frac{|b - b_n|}{|b| \cdot |b|/2} = \frac{|b - b_n|}{b^2/2}.$$

Now $b_n \rightarrow b$ and $\epsilon b^2/2 > 0$, so there is some $N_2 \in \mathbb{N}$ such that for all $n > N_2$,

$$|b - b_n| < \epsilon b^2/2.$$

Put $N = \max\{N_1, N_2\}$. Then for all $n > N$,

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| < \frac{|b - b_n|}{b^2/2} < \frac{\epsilon b^2/2}{b^2/2} = \epsilon.$$

■

From these rules, we can get many more useful results:

Proposition 6.3 Suppose that $x_n \rightarrow L$ as $n \rightarrow \infty$ and that $k \in \mathbb{N}$. Then $x_n^k \rightarrow L^k$ as $n \rightarrow \infty$.

Proof. We prove this by induction on k . The base case $k = 1$ is a tautology. For the induction step, assume we have shown that whenever $x_n \rightarrow L$, then for some particular $k \geq 1$, $x_n^k \rightarrow L^k$. Now apply Proposition 6.2(b) with $a_n = x_n^k$ and $b_n = x_n$, to obtain

$$\lim_{n \rightarrow \infty} x_n^{k+1} = \lim_{n \rightarrow \infty} x_n^k x_n = \left(\lim_{n \rightarrow \infty} x_n^k \right) \cdot \left(\lim_{n \rightarrow \infty} x_n \right) = L^k \cdot L = L^{k+1}$$

Thus, the proposition follows by induction. ■

Proposition 6.4 Let (x_n) and (y_n) be sequences.

- (a) If $x_n \rightarrow x$, $y_n \rightarrow y$, and $x_n \leq y_n$ for all $n \in \mathbb{N}$, then $x \leq y$.
 (b) If $x_n \rightarrow x$ and $x_n \leq y$ for all n , then $x \leq y$.

Proof. (a) Suppose for a contradiction that $x > y$. Take $\epsilon = (x - y)/2 > 0$. Since $x_n \rightarrow x$, there is N_1 such that $n > N_1$ implies

$$x - \frac{x - y}{2} < x_n < x + \frac{x - y}{2}$$

and in particular

$$\frac{x + y}{2} < x_n.$$

Similarly, since $y_n \rightarrow y$, there is N_2 such that $n > N_2$ implies

$$y - \frac{x - y}{2} < y_n < y + \frac{x - y}{2}$$

and in particular

$$y_n < \frac{x + y}{2}.$$

Therefore, for all $n > \max\{N_1, N_2\}$,

$$\frac{x + y}{2} < x_n \leq y_n < \frac{x + y}{2}.$$

which gives the desired contradiction.

(b) This follows by (a) by letting $y_n = y$ for all n . ■

There is another important rule for limits, often called the "squeeze theorem":

Proposition 6.5 Suppose that $x_n \leq y_n \leq z_n$ for all n , and that $x_n \rightarrow L$ and $z_n \rightarrow L$. Then $y_n \rightarrow L$.

Proof. Let $\epsilon > 0$. There exist N_1 and N_2 such that for $n > N_1$ we have

$$|x_n - L| < \epsilon$$

and for $n > N_2$ we have

$$|z_n - L| < \epsilon.$$

Take $N = \max\{N_1, N_2\}$. For $n > N$ we have

$$-\epsilon < x_n - L \leq y_n - L \leq z_n - L < \epsilon.$$

Therefore for $n > N$ we have $|y_n - L| < \epsilon$. ■

6.3 Application: the existence of roots.

Recall that in Week 1 we asserted the existence of roots. Now we can now prove it.

Theorem 6.6 Let $x > 0$ and $k \in \mathbb{N}$. Then there is a unique $y > 0$ such that $y^k = x$.

Proof. Let $S = \{s \geq 0 \mid s^k < x\}$. Then S is nonempty (since $0 \in S$) and bounded above since $\max\{x, 1\}$ is an upper bound for S . Therefore

$$y = \text{LUB}(S) = \text{LUB}(\{s \geq 0 \mid s^k < x\})$$

exists by the Completeness Axiom. We claim that $y^k = x$.

Since y is the least upper bound of S , this means that for all $n \in \mathbb{N}$, $y - \frac{1}{n}$ is not an upper bound, i.e. there is $y_n \in S$ such that

$$y - \frac{1}{n} < y_n \leq y.$$

Now $y - \frac{1}{n} \rightarrow y$ (by Example 6.1 and the rules for limits), and the constant sequence $y, y, \dots$ converges to y too, so by the squeeze theorem, $y_n \rightarrow y$. By Proposition 6.3, $y_n^k \rightarrow y^k$. But $y_n \in S$, so $y_n^k < x$ for all n , so Proposition 6.4 gives

$$y^k \leq x.$$

Now we will show the reverse inequality, i.e. that $x \leq y^k$, and this will imply $y^k = x$. Since $y = \text{LUB}(S)$, y is an upper bound for S , and so for all n , $y + \frac{1}{n} \notin S$. By the definition of S , this means

$$\left(y + \frac{1}{n}\right)^k \geq x.$$

Again, Propositions 6.3 and 6.4, together with the fact that $y + \frac{1}{n} \rightarrow y$, imply that

$$y^k \geq x.$$

Thus, $y^k = x$.

Finally, if $0 < w < z$ then $w^k < z^k$ (see Proposition 4.3) and so there is a *unique* $y > 0$ such that $y^k = x$. ■

6.4 Infinite limits

Definition 6.5 Let (x_n) be a sequence of real numbers.

- (a) We say (x_n) *tends to* ∞ or *diverges to* ∞ , and write $x_n \rightarrow \infty$ as $n \rightarrow \infty$, if for all $M > 0$, there exists $N \in \mathbb{N}$ such that $n > N$ implies $x_n \geq M$.
- (b) Similarly, we say x_n *tends or diverges to* $-\infty$, and write $x_n \rightarrow -\infty$ as $n \rightarrow \infty$, if for all $M < 0$, there exists $N \in \mathbb{N}$ such that $n > N$ implies $x_n \leq M$.

Example 6.4 To show that $\sqrt{n} \rightarrow \infty$, let $M > 0$. We need to find $N \in \mathbb{N}$ such that $n > N$ implies $\sqrt{n} > M$. If we pick N to be an integer greater than M^2 , then for all $n > N$,

$$\sqrt{n} > \sqrt{N} > \sqrt{M^2} = M.$$

We have shown that for any $M > 0$ we can find $N \in \mathbb{N}$ such that $n > N$ implies $\sqrt{n} \geq M$. Hence $\sqrt{n} \rightarrow \infty$.

Example 6.5 The sequence $x_n = n^2 - n - 1$ tends to ∞ . To prove this, we must show that given $M > 0$, there is some $N \in \mathbb{N}$ such that for all $n > N$,

$$n^2 - n - 1 > M.$$

Note that for $n > 1$,

$$n^2 - n - 1 = n(n-1) - 1 \geq n \cdot 1 - 1 = n - 1$$

Thus, if N is an integer greater than $M + 1$, then for all $n > N$,

$$n^2 - n - 1 \geq n - 1 > N - 1 > M + 1 - 1 = M.$$

6.5 Exercises

The exercises in Liebeck's book relevant to this section are in Chapter 23.

■ **Exercise 6.1** Show that $|x - L| < \epsilon$ if and only if $L - \epsilon < x < L + \epsilon$.

■ **Exercise 6.2** Using the ϵ - N definition of a limit, prove the following:

- (a) $\lim_{n \rightarrow \infty} \frac{1}{n^3} = 0$.
- (b) $\lim_{n \rightarrow \infty} \frac{n^4}{n^4 + 1} = 1$
- (c) $\lim_{n \rightarrow \infty} \frac{n^2 + 3}{n^2 + n} = 1$
- (d) $\lim_{n \rightarrow \infty} \frac{2n^2 - n}{n^2 + n - 1} = 2$.

■ **Exercise 6.3** For the following sequences, show that they tend to infinity.

- (a) $5n - 2$
- (b) $n^3 + 1$
- (c) $n^4 - n + 1$
- (d) $n!$
- (e) $\sqrt{n}$

■ **Exercise 6.4** Show that the limit of a sequence is unique, that is, if $x_n \rightarrow L$ and $x_n \rightarrow M$, then $L = M$.

■ **Exercise 6.5** For each L below, find sequences (x_n) and (y_n) such that $x_n \rightarrow 0$, $y_n \rightarrow 0$ and $x_n \neq 0, y_n \neq 0$ for all n , and such that $\frac{x_n}{y_n} \rightarrow L$.

- (a) 1
- (b) 0
- (c) ∞

■ **Exercise 6.6** For each L below, find sequences (x_n) and (y_n) such that $x_n \rightarrow 0$, $y_n \rightarrow \infty$ and $x_n y_n \rightarrow L$.

- (a) 1
- (b) 0
- (c) ∞

■ **Exercise 6.7** Show that if $x_n \rightarrow x$, then $|x_n| \rightarrow |x|$.

■ **Exercise 6.8** Suppose that (x_n) converges. Show that for any $k \in \mathbb{N}$, $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_{n+k}$.

■ **Exercise 6.9** Suppose that (x_n) converges. Show that $x_n - x_{n+1} \rightarrow 0$.

■ **Exercise 6.10** Using the ϵ - N definition of a limit, prove that if $x_n > 0$ and $x_n \rightarrow x > 0$ and $x_n \neq x$ for all n , then $x_n^{\frac{1}{3}} \rightarrow x^{\frac{1}{3}}$.

- **Exercise 6.11** Let $a_n = \left(1 - \frac{1}{n^2}\right)^n$. Show that $a_n \rightarrow 1$ as $n \rightarrow \infty$. What happens if we consider $\left(1 - \frac{1}{n^s}\right)^n$ for $s > 1$? (**Hint:** Bernoulli's inequality (see workshop W02).)
- **Exercise 6.12** Let $a_n = \left(1 + \frac{1}{n^2}\right)^n$. Show that $a_n \rightarrow 1$ as $n \rightarrow \infty$.
- **Exercise 6.13** Let $a_n = \left(1 + \frac{1}{n}\right)^n$. Given that $a_n \rightarrow L$ as $n \rightarrow \infty$, what can you say about the sequence (b_n) where $b_n = \left(1 - \frac{1}{n}\right)^n$?